AI-Powered Campus Assistant (Campus Buddy)

AICTE Lenovo LEAP NextGen Scholar – Capstone Project

Student Name

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Project Title

AI-Powered Campus Assistant (Campus Buddy)

1. Selected SDG and Reason

Sustainable Development Goal (SDG) 4: Quality Education

Quality education is one of the key goals of the United Nations Sustainable Development Goals (SDGs). Educational institutions generate a large amount of information related to admissions, courses, fees, academic schedules, and student services. Students often face challenges in accessing this information quickly and accurately.

The AI-Powered Campus Assistant supports SDG 4 by providing students with instant access to essential academic information. The chatbot acts as a virtual assistant that can answer common student queries at any time, improving accessibility and promoting a better learning experience.

2. Problem Statement

In educational institutions, students frequently need information regarding admissions, courses, fee structures, college timings, and contact details. Traditionally, students obtain this information through websites, notice boards, or administrative offices. This process can be time-consuming and may lead to delays in obtaining the required information.

Many students also face difficulties when information is scattered across different sources. There is a need for a centralized and user-friendly solution that can provide accurate information instantly. Therefore, an AI-based chatbot can help students by providing quick responses and reducing dependency on manual support.

3. Project Objective

The primary objective of this project is to develop an AI-powered chatbot that serves as a virtual campus assistant for students.

The specific objectives are:

- To provide instant responses to student queries.
- To improve accessibility to academic information.
- To reduce the workload on administrative staff.
- To enhance user experience through automated interactions.
- To demonstrate the practical application of conversational AI in education.

4. Proposed Solution

The proposed solution is an AI-Powered Campus Assistant named Campus Buddy. The chatbot is developed using Botpress and follows a structured conversational workflow.

The chatbot interacts with users by displaying a welcome message and guiding them through various options. Users can ask questions related to admissions, courses, fees, college timings, and contact details. The chatbot then provides predefined responses based on the selected option.

A fallback mechanism is also included to handle queries that are not recognized by the system. This ensures that users receive appropriate guidance even when the chatbot cannot understand a specific request.

5. Project Features

Admission Information

Provides information about the admission process and required procedures.

Course Information

Displays details about available academic programs and courses offered by the institution.

Fee Information

Provides information regarding fee-related queries and payment details.

College Timings

Informs students about the operating hours of the institution.

Contact Information

Provides important contact details for student assistance.

User-Friendly Interface

Offers a simple and interactive interface for easy communication.

Fallback Response

Handles unknown or unsupported queries and guides users appropriately.

Automated Assistance

Provides instant responses without requiring human intervention.

6. Technology Stack

The following technologies were used in the development of this project:

Botpress

Botpress was used to design and develop the chatbot workflow.

Artificial Intelligence Concepts

Basic AI concepts such as conversational agents and automated response systems were applied.

GitHub

GitHub was used for project documentation and version control.

Web Browser

The chatbot was tested and executed through a web-based interface.

7. Working of the Project

The working of the chatbot follows a simple conversational process:

1. The user opens the chatbot interface.
2. The chatbot displays a welcome message.
3. The user selects or enters a query.
4. The chatbot processes the request using predefined conversation nodes.
5. The relevant response is displayed to the user.
6. If the query is not recognized, the chatbot provides a fallback response.
7. The interaction continues until the user receives the required information.

This workflow ensures smooth communication between the user and the chatbot.

8. Project Screenshots

Figure 1: Workflow of Campus Buddy Chatbot

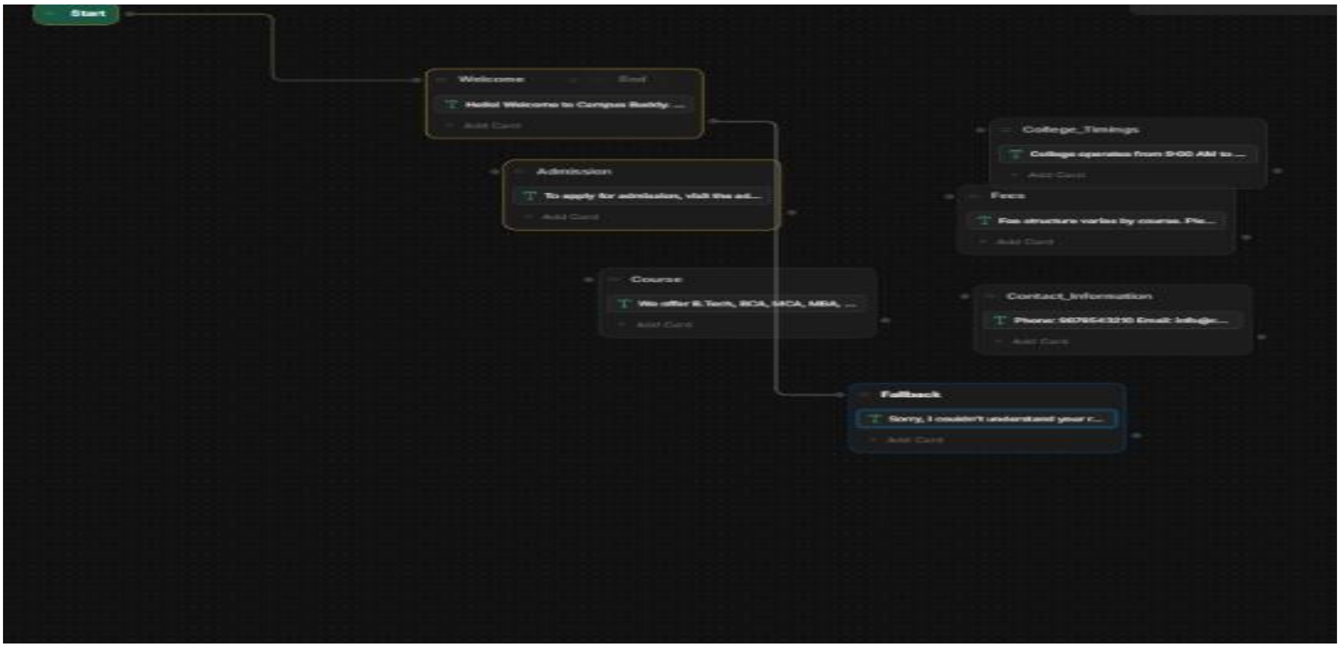


Figure 2: Welcome Screen

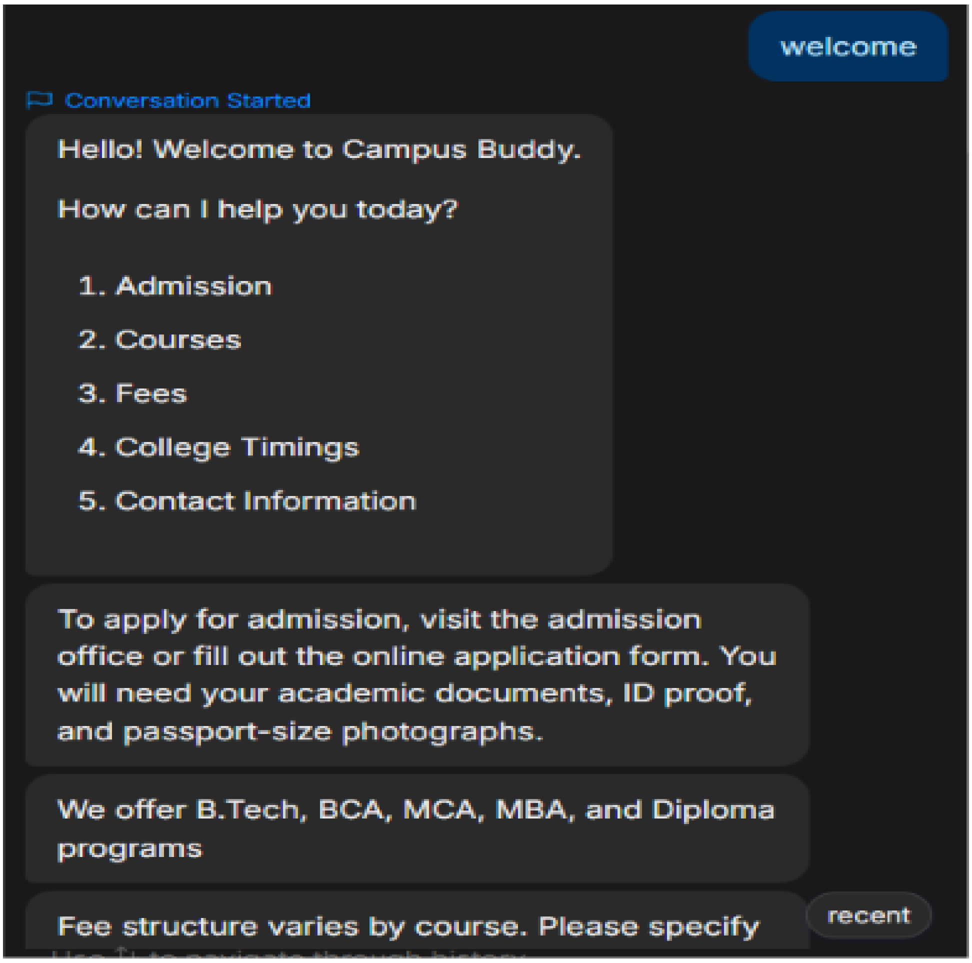


Figure 3: Admission Information Response

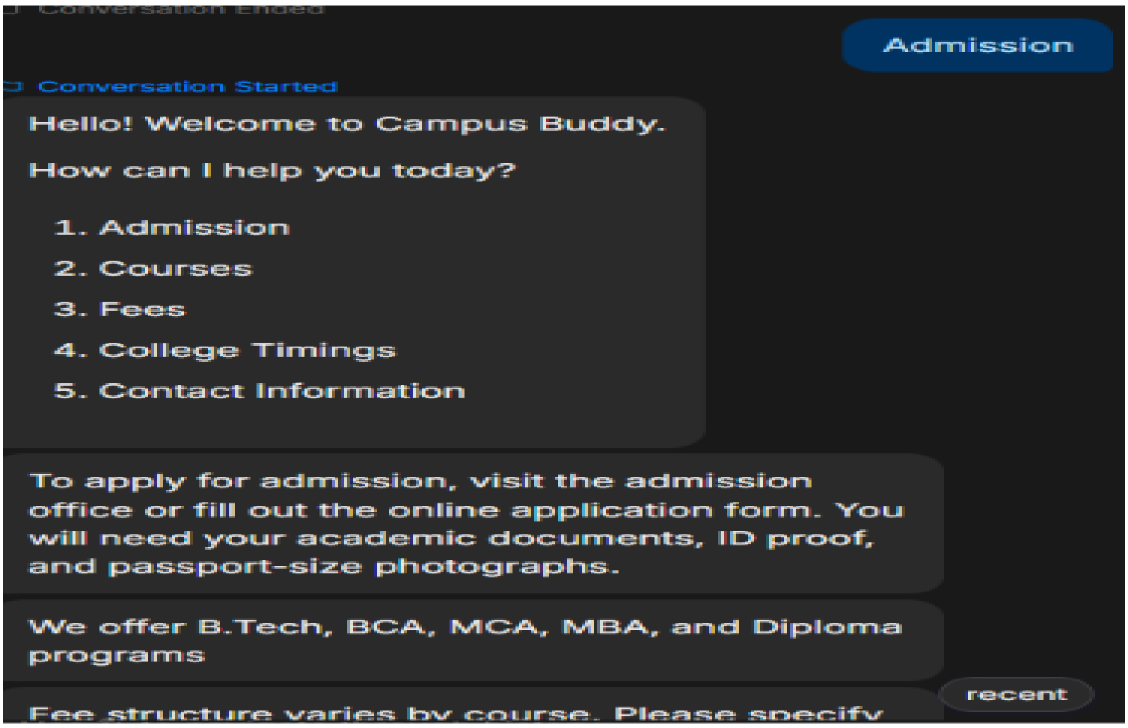


Figure 4: Course Information Response

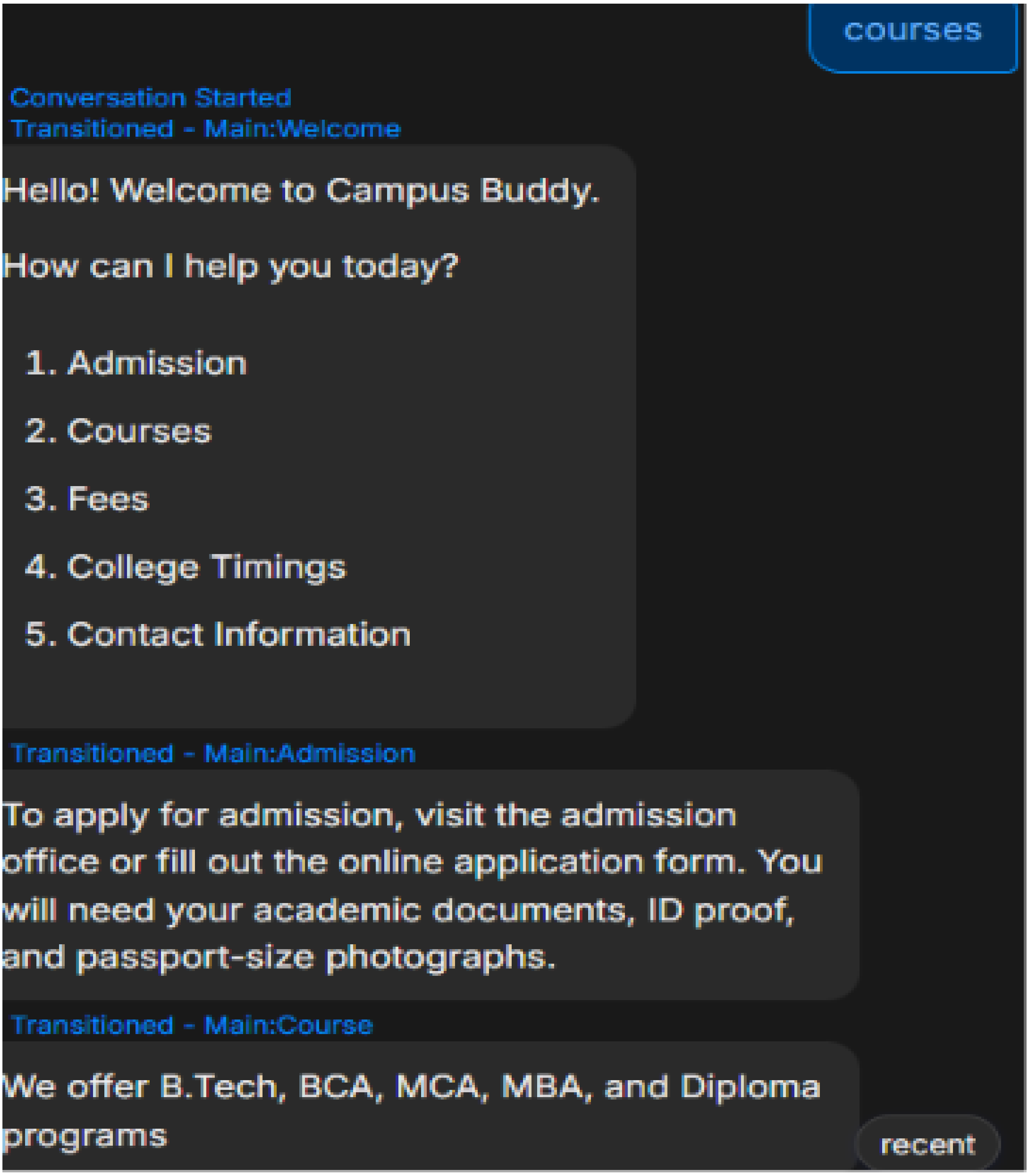
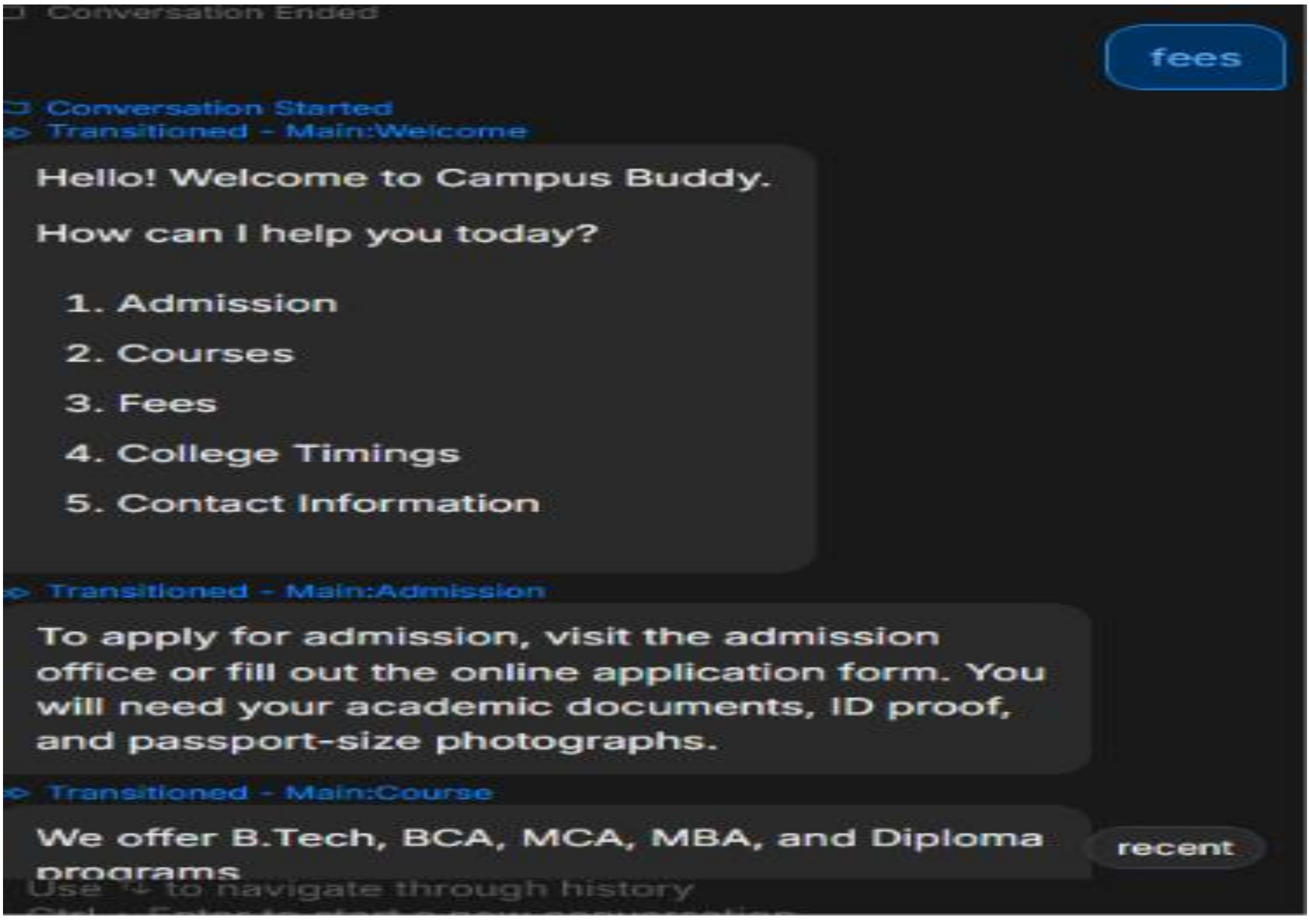


Figure 5: Fee Information Response


Conversation Ended
fees
Conversation Started
Transitioned - Main:Welcome
Hello! Welcome to Campus Buddy.
How can I help you today?
1. Admission
2. Courses
3. Fees
4. College Timings
5. Contact Information
Transitioned - Main:Admission
To apply for admission, visit the admission office or fill out the online application form. You will need your academic documents, ID proof, and passport-size photographs.
Transitioned - Main:Course
We offer B.Tech, BCA, MCA, MBA, and Diploma programs
recent
Use ↑↓ to navigate through history

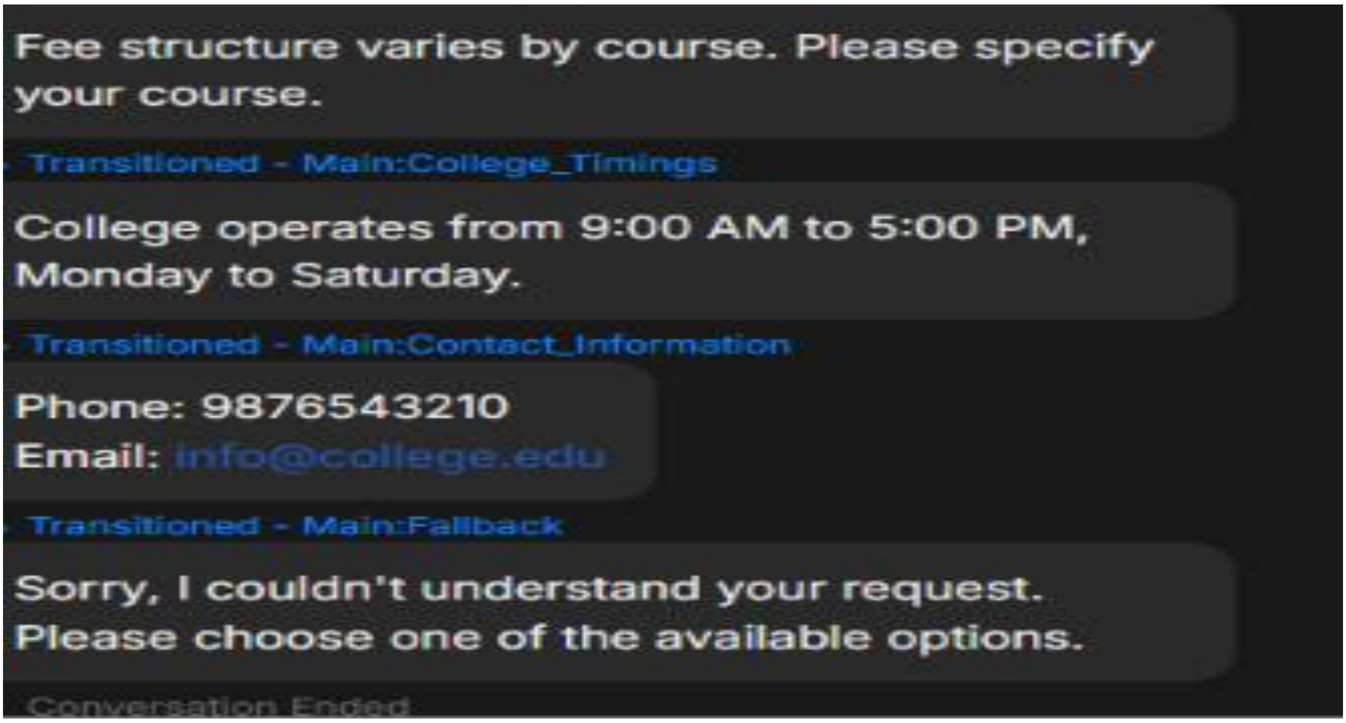
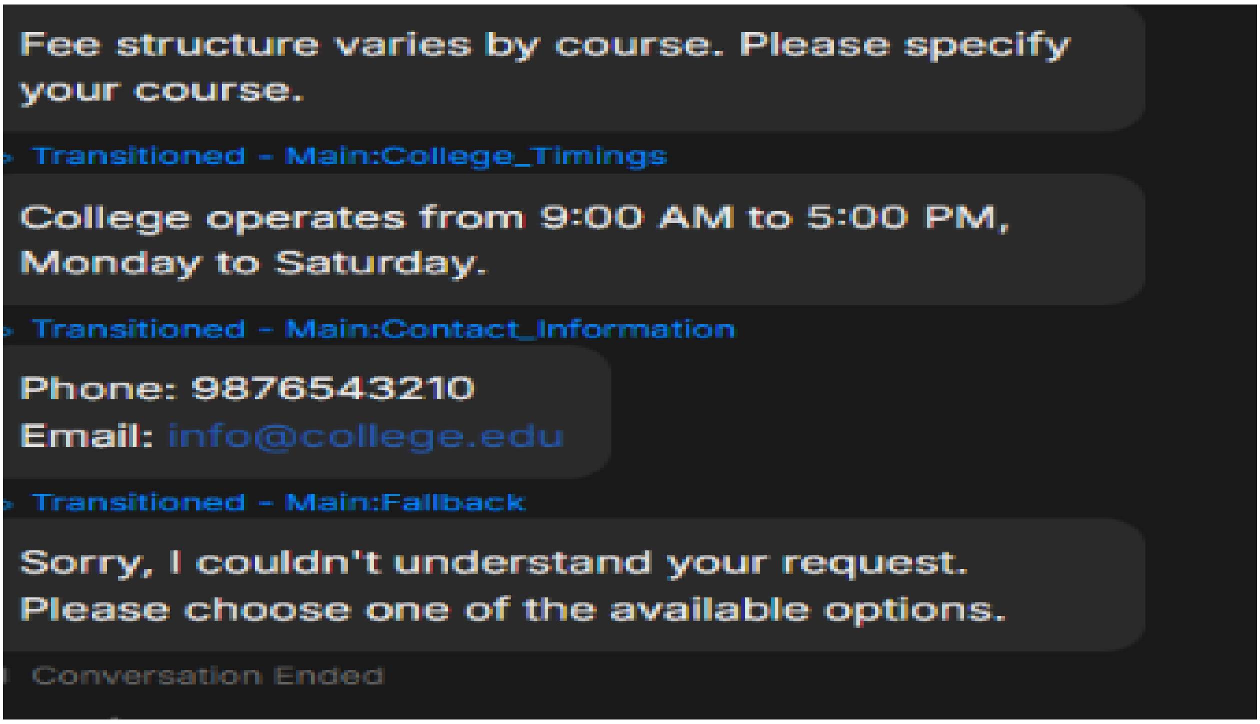

Fee structure varies by course. Please specify your course.
Transitioned - Main:College_Timings
College operates from 9:00 AM to 5:00 PM, Monday to Saturday.
Transitioned - Main:Contact_Information
Phone: 9876543210
Email: info@college.edu
Transitioned - Main:Fallback
Sorry, I couldn't understand your request. Please choose one of the available options.
Conversation Ended

Figure 6: Contact Information Response



9. Results and Outcomes

The chatbot was successfully developed and tested using Botpress. The system was able to provide accurate responses to predefined user queries related to admissions, courses, fees, college timings, and contact details.

The project demonstrated the effectiveness of conversational AI in improving access to educational information. Users were able to receive information quickly without visiting multiple sources or contacting administrative staff.

The chatbot achieved its objective of providing fast, automated, and user-friendly assistance.

10. Future Scope

Although the current version of the chatbot provides predefined responses, several improvements can be implemented in future versions.

Future enhancements include:

- Voice-enabled interaction
- Multilingual language support
- Integration with college databases
- AI-powered intelligent responses using Generative AI
- Mobile application integration

- Personalized student assistance
- Real-time notifications and updates

These enhancements can significantly improve the functionality and usefulness of the chatbot.

11. Conclusion

The AI-Powered Campus Assistant (Campus Buddy) is a practical application of conversational AI in the education sector. The project successfully addresses the challenge of providing quick access to academic information and demonstrates how chatbot technology can improve communication between students and institutions.

The project supports Sustainable Development Goal 4 (Quality Education) by making educational information more accessible, organized, and readily available. Through this project, valuable knowledge was gained in chatbot development, workflow design, AI concepts, and user interaction design.

Paste your GitHub repository link here:

https://github.com/shravani-1508/Campus-Buddy-Chatbot.git